

[B] AISSA – AI Software System Assistant team 4

1. Project Overview

The AI Software System Assistant (AISSA) is an embedded, AI driven digital secretary designed to be installed at an office entrance. The system provides human–machine interaction through a digital face displayed on a screen , combined with a camera and microphone to detect visitors, initiate interaction, collect messages, and provide information about the office owner.

When a person is detected by the camera, AISSA activates the digital face (e.g., opens its eyes) and greets the visitor with the phrase: “May I help you?” . AISSA then engages in a short conversational interaction to either:

Provide predefined information (office owner availability, contact details, office hours), or

Capture and store a message from the visitor.

2. Development Constraints and Effort

Team and Time Allocation

Team size: 6 members

Effort per developer: 5 hours/week

Total team effort: 30 hours/week

Development Cycles

Cycle 1: 16 days → 180 hours available

Cycle 2: 18 days → 210 hours available

Total project effort: 310 hours

3. Stakeholders and User Roles

- Office Owner
- Visitors
- System Administrator
- Maintenance / IT Support

User Roles

- Visitor: Interacts with AISSA to obtain information or leave a message
- Office Owner: Receives messages and configures information
- Administrator: Manages system configuration, security, and maintenance

4. Functional Requirements

4.1 Presence Detection and Activation

FR 01: The system shall continuously monitor the camera feed for human presence.

FR 02: Upon detecting a person, the system shall activate the digital face on the display.

FR 03: The digital face shall visually respond by opening its eyes.

FR 04: The system shall greet the detected person with the phrase “May I help you?”.

4.2 Human–AI Interaction

FR 05: The system shall support spoken interaction using speech recognition.

FR 06: The system shall display facial expressions synchronized with speech output.

FR 07: The system shall respond to basic conversational intents (greeting, request for information, message leaving).

FR 08: The system shall handle interaction timeouts when no response is detected.

4.3 Message Capture and Management

FR 09: The system shall allow visitors to leave a spoken message.

FR 10: The system shall convert spoken messages to text.

FR 11: The system shall store messages with timestamp and optional image snapshot.

FR 12: The system shall notify the office owner of new messages.

4.4 Information Provision

FR 13: The system shall provide predefined information about the office owner (name, role, office hours).

FR 14: The system shall inform visitors of the owner's availability status.

FR 15: The system shall provide alternative contact information when the owner is unavailable.

4.5 Administration and Configuration

FR 16: The system shall allow configuration of greeting messages.

FR 17: The system shall allow updating office owner information.

FR 18: The system shall provide access control for administrative functions.

4.6 Logging and Monitoring

FR 19: The system shall log all interactions.

FR 20: The system shall store system events for diagnostics.

5. Non Functional Requirements

5.1 Performance

NFR 01: Presence detection shall occur within 1 second.

NFR 02: Speech response latency shall be less than 2 seconds.

5.2 Reliability and Availability

NFR 03: The system shall operate continuously during office hours.

NFR 04: The system shall recover automatically from minor failures.

5.3 Security and Privacy

NFR 05: All stored messages shall be encrypted at rest.

NFR 06: Audio and image data shall comply with privacy regulations.

NFR 07: Administrative access shall require authentication.

5.4 Usability

NFR 08: The interaction shall be intuitive for first time users.

NFR 09: The digital face shall clearly convey active and idle states.

5.5 Maintainability

NFR 10: The system shall be modular and allow component updates.

NFR 11: The system shall include diagnostic and logging capabilities.

5.6 Portability

NFR 12: The system shall run on embedded hardware platforms.

6. Data Requirements

- Visitor interaction logs
- Voice recordings (optional, configurable)
- Transcribed text messages
- System status and error logs

7. Technology Stack (Recommended)

7.1 Hardware

- Embedded system (e.g., NVIDIA Jetson Nano / Raspberry Pi 5)
- HD camera
- Microphone and speaker
- Touch or standard display

7.2 Software Architecture

- Embedded Linux OS
- Modular, service oriented architecture

7.3 AI and Interaction

- Face detection: OpenCV / MediaPipe
- Speech recognition: Whisper / Google Speech API
- Text to speech: Coqui TTS / Amazon Polly
- Conversational logic: LLM based assistant (local or cloud)

7.4 Frontend

- Digital face rendering: Unity / Three.js
- UI control via Python or C++ bindings

7.5 Backend

- Language: Python
- Local database: SQLite / PostgreSQL
- Messaging/notification: Email or mobile app integration

8. Development Plan (PMI Aligned)

Cycle 1 (180 hours): Core AISSA

- Project initiation and scope definition
- Requirements and WBS
- Presence detection and greeting
- Digital face rendering
- Basic speech input/output
- Testing

Cycle 2 (210 hours): Advanced Interaction

- Message storage and notification
- AI conversational refinement
- Security and privacy controls

- System integration testing
- User acceptance testing

9. PMI Project Management Application

- Integration Management
- Project charter and integrated change control
- Scope Management
- Scope baseline using WBS
- Schedule Management
- Time boxed cycles with milestones
- Cost Management
- Effort based cost estimation
- Quality Management
- Acceptance criteria and testing plan
- Risk Management
- Risks: privacy, AI accuracy, hardware reliability
- Communications Management
- Weekly status reports and demos
- Stakeholder Management
- Continuous feedback from office owner